Daniel Costa Araújo

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Research Interests

Statistical and array signal processing; parameter estimation in propagation channels; massive MIMO and MIMO-OFDM channel estimation; hybrid beamforming; localization and integrated sensing and communication (ISAC) for 5G/6G; machine learning for the physical layer; software-defined radio and microwave systems.

Education

- **Federal University of Ceará (UFC)** — Fortaleza, Brazil
  Postdoctoral Fellow in Teleinformatics Engineering – CNPq — *2017 – 2018*
- **Federal University of Ceará (UFC)** — Fortaleza, Brazil
  Ph.D. in Teleinformatics Engineering (CAPES/PDSE sandwich at Ericsson Research) — *Aug. 2012 – Sep. 2016*
  - **Thesis**: Channel Estimation Techniques Applied to Massive MIMO Systems Using Sparsity and Statistical Approaches
  - **Advisors**: J. C. M. Mota (UFC); A. L. F. de Almeida (UFC, co-advisor); E. Karipidis (Ericsson)
- **Federal University of Ceará (UFC)** — Fortaleza, Brazil
  M.Sc. in Teleinformatics Engineering – CAPES Scholarship — *Aug. 2010 – Jun. 2012*
- **University of Fortaleza (UNIFOR)** — Fortaleza, Brazil
  B.Sc. in Telecommunications Engineering — *Aug. 2005 – Jul. 2010*

Academic Positions

- **University of Brasília (UnB) – FCTE, Gama campus** — Brasília, DF, Brazil
  Adjunct Professor — *December 2019 – Present*
  - **Coordinator, B.Sc. in Electronic Engineering**: September 2023 – 2025
  - **Deputy Coordinator, B.Sc. in Electronic Engineering**: April 2020 – April 2023
  - **Vice-Coordinator**: LabTelecom – Shared Teaching and Research Laboratory in Telecommunications (LCEPT)
- **UFC – GTEL / Ericsson Research Program** — Fortaleza, Brazil
  Research Engineer (Wireless Telecommunications Research Group) — *Jan. 2012 – Dec. 2019*
- **Ericsson Research AB** — Kista, Sweden
  Visiting Researcher – RAT (Radio Access Technology) Department — *May 2015 – Apr. 2016*
- **Centro Universitário Christus / 7 de Setembro** — Fortaleza, Brazil
  Lecturer (Applied Math, Probability & Statistics, Digital Electronics, Linear Systems) — *Feb. 2017 – Jun. 2019*

Funded Research Projects (Principal Investigator)

- **FAPDF – 6G Localization & Sensing Platform**: Development testbed for angular profile estimation and waveform design for 6G ISAC (2022–2024)
- **FAPDF – Low-cost LEO satellite Internet platform**: Holographic and reflectarray antennas with ML-based beam steering for Starlink-class downlink (2022–2024)
- **CNPq – UNIESPAÇO 2022**: Distributed low-cost ground station with antenna arrays for LEO satellite downlink (2023–Present)
- **Industry partnerships**: Tecsys Industrial – SDR implementation of SIGFOX and LoRa protocols (2023–2024); SoC-FPGA digital TV system (2024)
- **SEMFOGO-DF**: AI/Computer Vision system for wildfire detection in the Brazilian Cerrado (2022–2024, with CBM-DF)

Teaching (UnB)

- **Graduate (PPGEE)**: Adaptive Array Signal Processing
- **Undergraduate (Electronic Eng.)**: Communication Principles for Engineering; Signal Integrity in Circuit Design; Practical Electronic Circuits I

Student Supervision

- **M.Sc. Theses (principal advisor, completed)**: 3 – L. A. A. Rocha (2021, near-field user localization), A. B. Bigonha (2023, cell-free MIMO clustering), G. B. Pinheiro (2025, adaptive array signal processing for LEO)
- **Ongoing**: 1 Ph.D. (A. C. P. Silva, coexistence studies in IMT systems) and 3 M.Sc. (B. C. Teixeira, UAV localization via distributed ISAC; H. O. Silva, lens-based DOA estimation and 77 GHz radar; K. Kuster, LEO orbit tracking with ground-station antenna arrays)
- **Ph.D. Thesis (co-advisor)**: F. H. Costa Neto (UFC, 2020, ML-based radio resource management for 5G)
- **Undergraduate Research (IC) and Final Projects (TCC)**: Numerous projects in sparse GNSS signal processing, cell-free MIMO beamforming, holographic antennas, Doppler estimation for LoRa-satellite links, and Butler matrices for directional beamforming
- **M.Sc. Examining Committees**: Member of 10+ M.Sc. defense committees in Electrical and Teleinformatics Engineering (UnB, UFC)

Awards

- **SBrT 2026 Best Paper Award**: Brazilian Telecommunications Society (SBrT), Full Papers, Telecommunications category, for the paper "Channel Estimation for Flexible Intelligent Metasurface Aided MIMO Communications" (SBrT 2026)
- **IET Communications Premium Award (2018)**: Institution of Engineering and Technology, for the paper "Massive MIMO: Survey and Future Research Topics" (IET Communications, 2016)

Publications (Summary)

- **Journal Articles**: 12 papers, including IEEE Wireless Communications Letters (3, 2026), IEEE Trans. Vehicular Technology, IEEE Access, IET Communications, Wireless Networks, Computer Communications, and Int. J. of Communication Systems
- **Conference Papers**: 28 papers in ICASSP, EUSIPCO, CAMSAP, LATINCOM, ITS, GLOBECOM Workshops, ISWCS, WCNPS, and SBrT
- **Patents**: 2 (WIPO international filing on RL-based uplink power control, WO 2021/223865; antenna beam control, US 2015/0264583)
- **Registered Software**: 1 – Computational Modeling for Indoor Localization in Industrial Environments (INPI BR 51 2026 001765-3, 2026)

Editorial and Professional Service

- **Associate Editor**: IEEE Signal Processing Letters (2025 – present)
- **Associate Editor**: Signal Processing Open (Elsevier), 2026 – present
- **Session Chair**: Technical session "Redes 5G/6G II" (5G/6G Networks II), SBrT 2026, Salvador, Brazil
- **Reviewer**: IEEE Trans. Communications, IEEE Wireless Communications Letters, Signal Processing Open, Circuits, Systems, and Signal Processing

Languages

Portuguese (native) | English (fluent) | Spanish (intermediate reading) | French (basic).